

HW3 Solutions

Math 742

2026-02-10

Casella/Berger - Problems

7.1 One observation is taken on a discrete random variable X with pmf $f(x|\theta)$, where $\theta \in 1, 2, 3$. Find the MLE of θ .

x	$f(x 1)$	$f(x 2)$	$f(x 3)$
0	1/3	1/4	0
1	1/3	1/4	0
2	0	1/4	1/4
3	1/6	1/4	1/2
4	1/6	0	1/4

MLE rule:

$$\hat{\theta}(x) = \begin{cases} 1 & \text{if } x = 0 \text{ or } x = 1 \\ 2 \text{ or } 3 & \text{if } x = 2 \text{ the likelihoods are tied} \\ 3 & \text{if } x = 3 \text{ or } 4 \end{cases}$$

7.11 Let $X_1, X_2, \dots, X_n$ be iid with pdf

$$f(x|\theta) = \theta x^{\theta-1}, 0 \leq x \leq 1, 0 < \theta < \infty$$

(a) Find the MLE of θ , and show that its variance $\rightarrow 0$ as $n \rightarrow \infty$

$$L(\theta; \mathbf{x}) = \prod_{i=1}^n \theta x_i^{\theta-1} = \theta^n \prod_{i=1}^n x_i^{\theta-1}.$$

$$\ell(\theta) = \log L(\theta) = n \log \theta + (\theta - 1) \sum_{i=1}^n \log x_i.$$

Differentiate and set to zero

$$\ell'(\theta) = \frac{n}{\theta} + \sum_{i=1}^n \log x_i = 0 \implies \hat{\theta} = -\frac{n}{\sum_{i=1}^n \log X_i}.$$

Show that $V(\hat{\theta}) \rightarrow 0$

Let $Y_i = -\log X_i$. Under this model, $Y_i \sim \text{Exp}(\text{rate} = \theta)$, so

$$S = \sum_{i=1}^n Y_i = -\sum_{i=1}^n \log X_i \sim \text{Gamma}(n, \text{rate} = \theta),$$

and hence

$$\hat{\theta} = \frac{n}{S}.$$

For $S \sim \text{Gamma}(n, \text{rate} = \theta)$ (with $n > 2$),

$$\mathbb{E}\left[\frac{1}{S}\right] = \frac{\theta}{n-1}, \quad \mathbb{E}\left[\frac{1}{S^2}\right] = \frac{\theta^2}{(n-1)(n-2)}.$$

Thus,

$$\mathbb{V}\left(\frac{1}{S}\right) = \mathbb{E}\left[\frac{1}{S^2}\right] - \left(\mathbb{E}\left[\frac{1}{S}\right]\right)^2 = \frac{\theta^2}{(n-1)(n-2)} - \left(\frac{\theta}{n-1}\right)^2 = \frac{\theta^2}{(n-1)^2(n-2)}.$$

Multiplying by n^2 ,

$$\text{Var}(\hat{\theta}) = n^2 \text{Var}\left(\frac{1}{S}\right) = \frac{n^2 \theta^2}{(n-1)^2(n-2)} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

(b) Find the method of moments estimator of θ .

For this distribution, $\mathbb{E}[X] = \frac{\theta}{\theta+1}$. Set

$$\bar{X} = \frac{\hat{\theta}_{MM}}{\hat{\theta}_{MM} + 1} \implies \hat{\theta}_{MM} = \frac{\bar{X}}{1 - \bar{X}}$$

Wasserman Section 9.14 - Problems

2. Let $X_1, X_2, \dots, X_n \sim \text{Uniform}(a, b)$, where a and b are unknown parameters and $a < b$.

(a) Find the method of moments estimators for a and b .

$$\mathbb{E}[X] = \frac{a+b}{2},$$

$$\mathbb{V}(X) = \frac{(b-a)^2}{12}.$$

Let $m_1 = \bar{X}$ and $m_2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$. Then

$$\hat{a}_{MM} = \bar{X} - \sqrt{3} \sqrt{m_2},$$

$$\hat{b}_{MM} = \bar{X} + \sqrt{3} \sqrt{m_2}.$$

(b) Find the MLE estimates of a and b .

The joint density is

$$f(\mathbf{x} \mid a, b) = (b-a)^{-n} \mathbf{1}_{\{a \leq X_{(1)}, X_{(n)} \leq b\}}.$$

Thus

$$\hat{a} = X_{(1)}, \quad \hat{b} = X_{(n)}.$$

(c) Let $\tau = \int x dF(x)$. Find the MLE of τ .

Let $\tau = \int x dF(x) = \mathbb{E}[X] = \frac{a+b}{2}$.

By invariance,

$$\hat{\tau} = \frac{\hat{a} + \hat{b}}{2} = \frac{X_{(1)} + X_{(n)}}{2}.$$

- (d) Let $\hat{\tau}$ be the MLE of τ . Let $\tilde{\tau}$ be the non-parametric plug-in estimator of $\tau = \int x dF(x)$. Suppose that $a = 1, b = 3$, and $n = 10$. Find the MSE of $\hat{\tau}$ by simulation. Find the MSER of $\tilde{\tau}$ analytically and compare to the simulated value.

$a = 1, b = 3, n = 10$. True $\tau = 2$.

- $\hat{\tau} = \text{MLE} = \frac{X_{(1)} + X_{(n)}}{2}$
- $\tilde{\tau} = \text{nonparametric plug-in} = \bar{X}$

Analytical MSE of $\tilde{\tau} = \bar{X}$:

$$\mathbb{V}(X_i) = \frac{(3-1)^2}{12} = \frac{1}{3},$$

$$\mathbb{V}(\bar{X}) = \frac{1/3}{10} = \frac{1}{30} \approx 0.03333.$$

Bias = 0 $\implies$ MSE($\tilde{\tau}$) = $\frac{1}{30}$.

```
set.seed(742)
n <- 10
a <- 1; b <- 3
tau_true <- (a + b)/2

B <- 50000
tau_hat <- numeric(B)

for(i in 1:B) {
  x <- runif(n, a, b)
  tau_hat[i] <- (min(x) + max(x))/2
}

mse_hat <- mean( (tau_hat - tau_true)^2 )
bias_hat <- mean(tau_hat) - tau_true
var_hat <- var(tau_hat)

c(MSE = mse_hat, Bias = bias_hat, Variance = var_hat)
```

```
##           MSE           Bias           Variance
## 0.0151936349 -0.0006812065 0.0151934747
```

- $\text{MSE}(\hat{\tau}) \approx 0.015$ (much smaller than $1/30 \approx 0.0333$)
- The MLE has lower MSE than the sample mean in small samples for uniform distribution.

6a. Let $X_1, X_2, \dots, X_n \sim N(\theta, 1)$. Define,

$$Y_i = \begin{cases} 1 & \text{if } X_i > 0 \\ 0 & \text{if } X_i \leq 0 \end{cases}$$

Let $\psi = \mathbb{P}(Y_1 = 1)$

(a) Find the MLE for ψ .

The Y_i are i.i.d. Bernoulli(ψ), so the MLE is

$$\hat{\psi} = \bar{Y} = \frac{1}{n} \sum Y_i = \text{proportion of } X_i > 0$$

By invariance, this is also the MLE of $\psi(\theta)$.

Problem 5. Determine the MoM for the Weibull distribution whose PDF is:

Shape $k > 0$ and scale $\lambda > 0$ is

$$f(x | k, \lambda) = \begin{cases} \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} \exp\left[-\left(\frac{x}{\lambda}\right)^k\right], & x > 0, \\ 0, & x \leq 0. \end{cases}$$

The first two moments are

$$\mathbb{E}[X] = \lambda \Gamma\left(1 + \frac{1}{k}\right), \quad \mathbb{E}[X^2] = \lambda^2 \Gamma\left(1 + \frac{2}{k}\right).$$

Let $m_1 = \bar{X}$ and $m_2 = n^{-1} \sum X_i^2$. Equating sample moments to population moments gives

$$m_1 = \hat{\lambda} \Gamma\left(1 + \frac{1}{\hat{k}}\right), \quad m_2 = \hat{\lambda}^2 \Gamma\left(1 + \frac{2}{\hat{k}}\right).$$

Dividing the second by the square of the first yields

$$\frac{m_2}{m_1^2} = \frac{\Gamma\left(1 + 2/\hat{k}\right)}{\left[\Gamma\left(1 + 1/\hat{k}\right)\right]^2}.$$

Let $c = m_2/m_1^2$. Solve numerically for $\hat{k}$ in

$$\frac{\Gamma\left(1 + 2/\hat{k}\right)}{\left[\Gamma\left(1 + 1/\hat{k}\right)\right]^2} = c,$$

then compute

$$\hat{\lambda} = \frac{m_1}{\Gamma\left(1 + 1/\hat{k}\right)}.$$

Numerical root-finding is usually required (e.g., `uniroot` in R).

Problem 6

1. Likelihood and log-likelihood

Because the sample is iid normal, the likelihood is

$$L(\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right).$$

Taking logs,

$$\ell(\mu) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

2. Show that maximizing the log-likelihood is equivalent to minimizing empirical MSE

The first term in $\ell(\mu)$ does **not** depend on μ . Thus maximizing $\ell(\mu)$ is equivalent to maximizing

$$-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

Because the constant $-\frac{1}{2\sigma^2} < 0$, maximizing this expression is equivalent to **minimizing** the quantity

$$\sum_{i=1}^n (x_i - \mu)^2.$$

Dividing by n (another positive constant) does not change the minimizer, so this is equivalent to minimizing the **empirical MSE**

$$\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2.$$

3. MLE for μ

Differentiate the objective:

$$\frac{d}{d\mu} \sum_{i=1}^n (x_i - \mu)^2 = -2 \sum_{i=1}^n (x_i - \mu).$$

Set equal to zero:

$$-2 \sum_{i=1}^n (x_i - \hat{\mu}) = 0 \implies \sum_{i=1}^n x_i = n\hat{\mu}.$$

Thus,

$$\hat{\mu} = \bar{X}.$$

So the MLE of μ is the sample mean, and it is the value that **minimizes the empirical MSE**.